

Amsheed Nazeer

AI, Deep learning engineer

Aspiring AI professional currently pursuing a Master's in Artificial Intelligence at the University of East London, supported by a Diploma in AI and hands-on experience in machine learning, deep learning, and data analysis. Alongside my studies, I work as a freelance AI developer, building custom chatbots, expert systems, RAG-based applications, and intelligent automation tools for businesses. Skilled in developing advanced AI systems, including dataset preprocessing, vectorization, retrieval pipelines, LLM fine-tuning, and deployment of RAG and multimodal models for real-world use cases.

I bring strong leadership, problem-solving, and decision-making abilities, and I am motivated to apply my technical and analytical skills to deliver impactful AI-driven solutions within a collaborative corporate environment

SKILLS

- Python, SQL, Java, R
- Pandas, NumPy, Seaborn, Matplotlib
- Machine Learning, Deep Learning, NLP (BERT, LSTM, GRU, Transformers, TensorFlow, ANN, CNN, RNN, NLP)
- Apache Spark, Hadoop
- Retrieval-Augmented Generation (RAG), Multi-agent system, Agentic AI
- Vector database, Embeddings, Semantic Search, LoRA, PEFT concepts
- n8n Automation, AI Orchestration, API Integrations
- Tensorflow, Model Fine-Tuning, Prompt Engineering
- Azure/IBM cloud/IBM Watson Studio
- REST API, Docker, CI/CD

PROFESSIONAL EXPERIENCE

Freelance AI Developer

Self-Employed | Remote | 2024 – Present

- Developed custom AI chatbots, expert systems, and automation tools for clients.
- Built RAG-based applications, retrieval pipelines, and multimodal AI solutions.
- Fine-tuned LLMs and deployed production-ready AI systems.
- Build academic and portfolio AI/ML projects for students, providing technical support, implementation guidance, and project execution.
- Delivered end-to-end solutions including data preprocessing, vectorization, and integration with FastAPI.

Deep Learning Intern — Oxi Academy

Machine Learning | ANN | CNN | TensorFlow | NLP (Jan 2025 – July 2025)

- Improved image recognition accuracy by implementing CNNs and transfer learning techniques.
- Built and optimized LSTM- and BERT-based NLP models for sequence and text processing tasks.
- Performed model evaluation, hyperparameter tuning, and deployment using TensorFlow and PyTorch.

Machine Learning Intern — Verzeo

Python | Pandas | NumPy | Data Visualization | ML Algorithms (Dec 2023 – March 2024)

- Used Python ML libraries to build predictive models with supervised and unsupervised learning techniques.
- Performed data preprocessing, feature extraction, and exploratory analysis to improve model performance.

Python Developer (Intern) — NXT Logic Solution

(Python | Decision Tree (01 Oct 2022 – 30 Nov 2022))

- Developed a crime prediction module using Decision Tree algorithms, achieving 87% prediction accuracy on real-world datasets.

Data Visualization Intern — Tata (Forage Virtual Internship)

(Power BI, Excel)

- Cleaned and transformed a dataset of 50,000+ records in Excel for analytical processing.
- Built interactive dashboards in Power BI to visualize key business insights and trends.

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EDUCATION

Masters in Artificial Intelligence- University of East London

(Expected to complete before Sep-2026)

Diploma in Artificial Intelligence

May 2024- July 2025

Bachelor of Science in Computer Science -

Hindusthan college of Arts and Science

Sep 2021-May 2024

CERTIFICATIONS

- **Infosys:** Soft skills, Public speaking, leadership
- **Coursera:** Advance Learning Algorithms, Powerbi
- **IBMcep:** IBM Watson Studio, IBM Cloud, Tensorflow

ACHIEVEMENTS

- **IBM Hackathon Challenge:** Ranked among the Top 150 teams in India for building an AI-based solution.
- **Best Leader:** Awarded Best Leader – Infosys CSR Programme for excellence in leadership, initiative, and impact.
- **Placement Head (UG):** Led placement operations and coordinated between students, faculty, and recruiting organizations.
- **DocMentor.AI:** Designed an AI assistant for project documentation, **proposed for adoption by the university** during my Master's program.

Projects

DocMentor.AI – AI-Powered Academic Documentation Intelligence Platform

Python, FastAPI, Azure AI Foundry, RAG, LLMs, Azure OpenAI, FAISS, Explainable AI

- Designed and implemented an AI-powered platform that **converts extended LLM conversations**, blueprints, and code into structured academic documentation aligned with university guidelines.
- Built a guideline intelligence pipeline that parses unstructured university PDFs into machine-readable rules with provenance and confidence scoring.
- Developed an agent-based documentation composer that generates section-level academic guidance (Introduction, Methodology, Evaluation, etc.) grounded in project context and institutional requirements.
- Implemented **Retrieval-Augmented Generation (RAG)** over guidelines, project conversations, and documentation drafts to provide explainable, context-aware academic assistance.
- Created an automated compliance and feedback engine that detects missing sections, guideline deviations, and structural gaps in student reports.
- Integrated visual intelligence features including AI-generated workflow flowcharts and submission-ready documentation templates.
- Deployed a modular **FastAPI backend** and integrated **Azure AI Foundry** agents for scalable, production-ready inference.

AI Multi-Agent Automation System

Python, CrewAI / LangGraph, LLM APIs, Prompt Engineering

- Built a collaborative multi-agent AI system using frameworks such as CrewAI / LangGraph.
- Designed specialized agents including:
 - **Research Agent** for trend and data analysis
 - **Copywriter Agent** for content generation
 - **Art Director Agent** for image prompt creation (Stable Diffusion)
 - **Manager Agent** to coordinate tasks and assemble final outputs
- Enabled agents to **communicate**, delegate tasks, and produce a unified marketing campaign brief.
- Demonstrated real-world automation of complex workflows using agent orchestration.

Self-Correcting Enterprise RAG Search Engine

Python, Qdrant, FastAPI, LangChain, Prompt Engineering

- Designed and implemented a multi-stage RAG system to improve factual accuracy and relevance of LLM responses.
- Built a document retrieval layer followed by a **Guardrail LLM** to filter and validate retrieved context.
- Implemented a generation agent to produce answers strictly grounded in approved sources.
- Developed an **Evaluator LLM agent** to score final outputs for factual consistency against source documents.
- Improved answer reliability by introducing automated self-correction and evaluation loops.